MemoryWeave: A Staged AI Pipeline for Reminiscence and Cross-Memory Reflection

Bin Yu
Digital Life Group
Faculty of Digital Media and Creative Industries
Amsterdam University of Applied Sciences
Amsterdam, the Netherlands

Maurits van der Does de Willebois
University of Amsterdam
Amsterdam, the Netherlands

Somaya Ben Allouch
Digital Life Group, Amsterdam University of Applied Sciences
Digital Interaction Lab, University of Amsterdam
Amsterdam, the Netherlands

Abstract

In this paper, we present MemoryWeave, a staged AI pipeline for memory-based reflection grounded in the two-phase structure of reminiscence therapy: first revisiting individual memories, then integrating them into a coherent life narrative. Stage 1 generates personalized savoring prompts grounded in individual photo-text memory entries through multimodal context extraction. Stage 2 builds on these reflections by first identifying thematic, emotional, and narrative connections across memories, and then using a grounded multi-turn dialogue to guide users in making meaning across multiple memories. We describe the system architecture, the AI pipeline, and the key design considerations of MemoryWeave. A planned within-subjects evaluation will examine whether this staged approach fosters deeper reflection and greater personal insight than single-memory reflection alone.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**.

Keywords

Reminiscence, Memory-based reflection, Savoring, AI pipeline, Mental wellbeing

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1 Introduction

People increasingly save meaningful life experiences through photographs and personal notes, building personal archives that can serve as rich materials for reminiscence and self-reflective insight [5, 16]. While many digital tools and apps [10] have been developed to help users revisit these memories, recent advances in AI have opened new possibilities for reflective technologies that are more personalized, adaptive, and capable of supporting deeper meaning-making [11, 12], for instance, AI-powered journaling tools [9, 19], conversational agents for self-reflection [14], and systems grounded in established therapeutic frameworks such as reminiscence therapy [6, 13, 23], narrative therapy [4], and positive psychology interventions[27]. These systems leverage large language models to generate personalized prompts based on user-provided content such as photographs, diary entries, or conversational responses, making therapeutic reflection more accessible, adaptive, and naturally embedded in everyday life.

However, reflecting on individual memories is only part of how personal meaning and insight are constructed. In therapeutic practice, integrative reminiscence [3] distinguishes two phases: first, the guided revisiting of individual memories to explore their personal significance; and second, the integration of these memories into a coherent life narrative through connections across experiences. This staged process is further supported by autobiographical reasoning theory [7], which describes how individuals construct identity by connecting distant life episodes to one another and to the present self [15]. In clinical practice, this resembles how a therapist first helps a person sit with a specific memory and then gradually prompts reflection on how it relates to other moments in their life. Despite the growing landscape of AI-supported reflection tools [4, 6, 9, 11–13, 19, 23, 27], no existing system explicitly stages these two processes, first supporting single-memory reflection and then enabling structured cross-memory meaning-making.

In this paper, we present MemoryWeave, a staged AI pipeline that combines both processes. Stage 1 generates personalized savoring prompts grounded in individual photo-text memory entries, supporting single-memory reminiscence reflection. Stage 2 builds on this foundation through two technical components: resonance detection, which identifies thematic, emotional, and narrative connections across memories, and a grounded multi-turn dialogue that guides users through cross-memory meaning-making. By staging these two processes together, MemoryWeave aims to support mental wellbeing by helping users not only revisit meaningful experiences, but also discover connections across them, fostering deeper self-understanding and personal insight.

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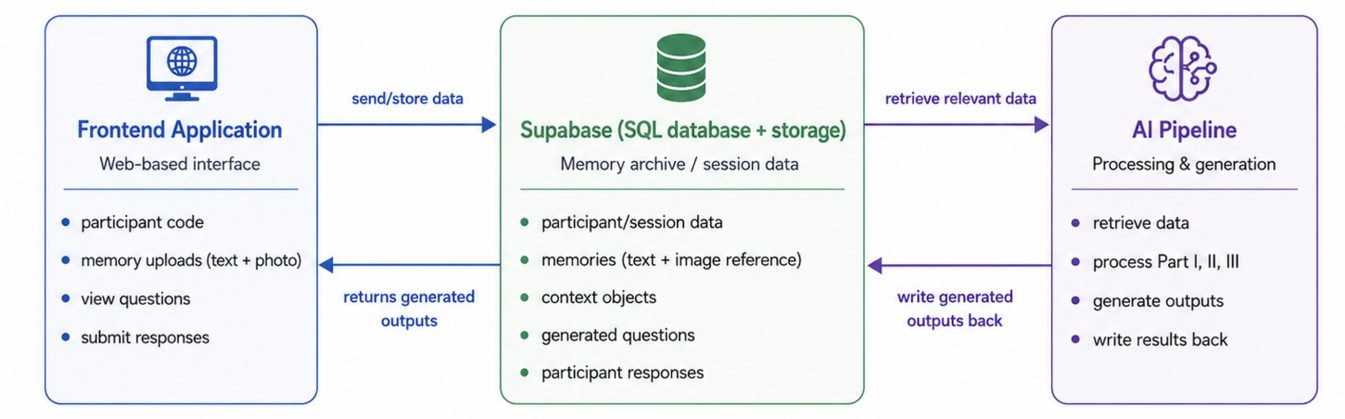


Figure 1: System overview of MemoryWeave application

2 Design

2.1 System overview

As shown in Fig 1, the system of MemoryWeave consists of three main components. The frontend allows users to upload personal photo-text memory entries at any time and provides the interface for reflection sessions. Each memory entry consists of a photograph and a written description, together capturing both the visual and personal significance of a memory. The backend consists of a database (Supabase) that persistently stores all memory entries, generated questions, and user responses, and an AI pipeline that retrieves memories from database, runs the staged processing on demand, and writes generated outputs back to the database.

2.2 User Interaction Flow

MemoryWeave supports two main forms of user interaction: memory upload and reflection sessions. In everyday life, users build their personal memory archive gradually by uploading photo-text memory entries at any time, for instance, capturing a meaningful moment shortly after it happens. When ready for a reflection session, such as in the evening, the system randomly selects six memories from the archive and initiates the two-stage pipeline. The selection of six memories per session was informed by our prior work [25], balancing session length (approximately 20–30 minutes) with the need to maintain user engagement while providing sufficient memory diversity for meaningful cross-memory meaning-making. In Stage 1, the six memories are presented to the user one by one, each accompanied by an AI-generated savoring prompt, and the user reflects on and responds to each memory individually. In Stage 2, the system uses the six memories and the user's reflection responses as the basis for a cross-memory dialogue, inviting the user to explore connections across their experiences in a multi-turn conversation.

3 The Two-Stage AI Pipeline

As shown in Fig. 2, the MemoryWeave AI pipeline is implemented as two sequential stages. In Stage 1, each memory is processed to extract a structured context object, which is then used to generate a personalized savouring prompt. In Stage 2, the six context objects and the user's Stage 1 reflection responses are passed through three parallel resonance detection sub-pipelines to identify cross-memory connections that seed a context-aware multi-turn dialogue. The details of each stage are presented in the following subsections.

3.1 Stage 1: Single-Memory Reflection

3.1.1 Multimodal Context Extraction. Each of the 6 selected memories consists of a photograph and a written description, which are jointly processed by a multimodal large language model (GPT-5.1) to produce a structured context object. Rather than processing the two modalities independently, the text serves as primary evidence for subjective memory qualities, while the image serves as primary evidence for visible and situational details, with cross-modal agreement only strengthening confidence when both sources clearly support the same detail. Ambiguous details are treated as unsupported rather than filled speculatively.

The context object captures two sets of dimensions: 10 memory dimensions grounded in the Memory Experiences Questionnaire (MEQ) [22], covering vividness, sensory detail, emotional valence, emotional intensity, visual perspective, time perspective, coherence, accessibility, sharing, and distancing; and 10 visual dimensions organized across three levels of abstraction [20, 21]: low-level dimensions (tone and color, texture and pattern, shape and form), spatial and scene-level dimensions (size and scale, depth and spatial cues, scene context), and semantic and compositional dimensions (semantic content, associations and function, composition, and visual hierarchy). Each extracted field carries a limited-evidence flag and confidence score, which propagate through the pipeline to constrain downstream reasoning to supported evidence only.

3.1.2 Savouring Prompt Generation. Using the extracted context object, the pipeline generates one personalized savouring prompt per memory. The prompt generation is grounded in savouring framework [1], which defines savouring as the capacity to attend to, appreciate, and enhance positive experiences. Specifically, the

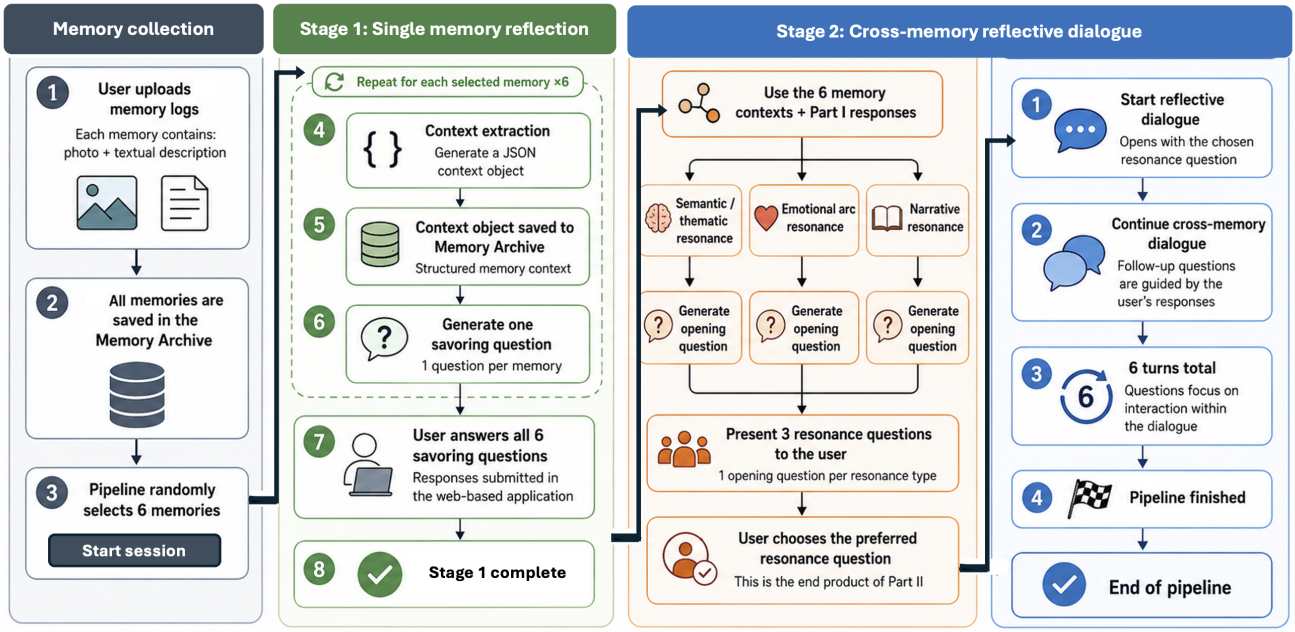


Figure 2: The detailed system workflow and two staged AI pipeline of MemoryWeave

pipeline focuses on the cognitive reflection dimension of savouring, which involves two processes: thanksgiving, an outward-focused reflection on gratitude toward others, fortune, or the external world; and basking, an inward-focused reflection on personal achievement, pride, and self-worth. The model is guided to classify each memory into one of these two processes based on the extracted context, select one grounded source of appreciation from the memory content, and generate exactly one open-ended, non-directive reflective question. The model is framed as an empathetic reflective-question generator inspired by Person-Centered Therapy principles [24], emphasizing non-directive support and self-understanding. Previous questions generated for other memories in the same session are also provided as input, allowing the model to avoid repetitive formulations across the six prompts.

3.2 Stage 2: Cross-Memory Meaning-Making

3.2.1 Resonance Detection. Once the user has responded to all six savoring prompts, Stage 2 begins. Rather than generating a generic cross-memory dialogue, the pipeline first identifies meaningful connections across the 6 memories and their reflection responses through three parallel resonance detection sub-pipelines. Each detected resonance type serves a dual purpose: it generates a grounded opening question that initiates the dialogue, and it provides the thematic anchor that guides the entire conversation, ensuring the dialogue remains rooted in the user's own material throughout.

The first type of resonance is thematic resonance. The pipeline identifies memories that share a meaningfully similar theme using their Stage 1 context objects as input. Each context object is first converted into a compact natural-language semantic digest, removing JSON field names and preserving only meaning-bearing content to prevent schema-induced similarity during embedding [26]. The digest captures thematic semantic fields informed by Mirman et al. [18], including event, situation, actions, relationships, and emotional qualities, before being embedded using text-embedding-3-large. This produces a 6x6 cosine similarity matrix representing the thematic similarity between all memory pairs. A candidate group is a subset of memories that share a meaningfully similar theme. Starting from the strongest memory pair, the pipeline expands each candidate group by including additional memories that keep the group mean similarity above the session mean. Each group is then scored by combining mean pairwise similarity, a size bonus, and an interpretability score. The top candidates are passed to an LLM interpretability check, which selects the single best grounded candidate or rejects all if none qualifies.

Secondly, for emotional resonance, the pipeline identifies recurring and contrasting emotional patterns across the six memories and their Stage 1 reflection responses. A recurring pattern occurs when similar emotional tones appear across multiple memories, while a contrasting pattern reflects clearly different emotional tones. The pipeline first infers an emotional register at three levels: the remembered event, the user's reflection response, and the emotional delta between them, capturing the dissociation between emotion at encoding and retrieval [8]. Emotions are represented both categorically, using a nine-label taxonomy (joy, sadness, anger, fear, disgust, surprise, calm/contentment, nostalgia/longing, pride) that extends basic emotions proposed in [2], and dimensionally using Valence–Arousal–Dominance (VAD) vectors [17], enabling complementary discrete and continuous representations. The taxonomy is designed to reduce label ambiguity while supporting autobiographically relevant states such as nostalgia and pride, and VAD enables

comparison across different labels in terms of valence, arousal, and agency. Recurring patterns are identified when multiple memories share dominant emotion labels or exhibit low VAD dispersion, while contrasting patterns are detected via distances between VAD centroids across event, reflection, and cross-level comparisons. All detected patterns are validated through an LLM-based interpretability check before being used for question generation.

Lastly, for narrative resonance, the pipeline classifies the six memories into one of five narrative types grounded in reminiscence therapy research [6, 13, 23]: belonging, growth, aspiration, loss, or resilience. A type is selected only when at least 4 of 6 memories explicitly support it, with strict rejection rules preventing spurious classification, for instance, memories are not classified as growth merely because they are positive, or as belonging merely because they include other people. When no type reaches the threshold the pipeline returns none, and when two types are equally supported it returns ambiguous. Each successfully detected resonance type generates one opening question grounded in the identified connection.

3.2.2 Grounded Multi-Turn Dialogue. Once the three resonance detection sub-pipelines have completed, each detected resonance type is used to generate an opening question via a shared prompt structured around Role, Setup, Task, and Output. Specifically, the model is instructed to act as an empathetic, non-directive reflective question generator grounded in Person-Centered Therapy principles [24], ensuring that each question is strictly based on the detected resonance and memory content while avoiding diagnostic language, unsupported interpretation, or technical terminology. The user is then presented with up to three candidate opening questions and selects one to start a reflective dialogue. The selected resonance initiates a context-aware multi-turn dialogue of up to six turns, where each subsequent question is conditioned on a four-part context window comprising the resonance memories and their context objects, the detected resonance connection, the full dialogue history, and the user's Stage 1 reflection responses. As the model does not retain memory across API calls, this context is re-supplied at each turn with an updated history to ensure continuity. At each step, the model follows a structured internal strategy by selecting one follow-up mode, such as deepening, clarifying, connecting, exploring surprise, inviting reinterpretation, broadening, or integrating—before generating a grounded, non-repetitive, autonomy-preserving reflective question.

3.3 Model Implementation and User Privacy

MemoryWeave processes sensitive personal data, including photographs and autobiographical memory descriptions, making privacy a first-order design requirement. All participant data and generated outputs are stored persistently in a project database within the university's Azure environment. For each pipeline step, only task-relevant data required for that specific processing stage is retrieved from the database and passed through a LiteLLM access layer to the model endpoint, after which the generated output is written back to the database. This design minimizes data exposure at each step and ensures that the model endpoint receives only the information necessary for the task. Model calls are routed through SURF's institutional AI Hub rather than a public consumer-facing interface, providing an additional layer of institutional oversight. GPT-5.1 and text-embedding-3-large are used as the primary language and embedding models, respectively, selected for their suitability for multimodal reasoning and semantic similarity tasks within the pipeline.

4 Discussion and and Future work

MemoryWeave proposes a novel staged approach to AI-supported memory-based reflection, combining single-memory reminiscence with cross-memory meaning-making in one pipeline. While existing AI reflection tools support either individual memory prompting or cross-memory connections in isolation, MemoryWeave is the first system to stage these two processes together: first inviting users to revisit individual memories, and then facilitating structured dialogue grounded in the connections across them.

This design was directly motivated by our prior work, MindFeed [25], an AI-assisted reflective journaling system that generated 6 personalized prompts by mixing multiple photo-text memory entries together. Although we hypothesized that AI-generated personalized prompts grounded in personal content would enhance reflection, the quantitative results did not support this hypothesis. The AI-generated prompts matched traditional journaling tools in engagement but varied in their ability to elicit deep personal insight. We observed that users felt confused when confronted with cross-memory connections without first having the opportunity to revisit each memory individually. This suggested that meaningful cross-memory meaning-making requires a prior stage of single-memory reflection to prepare users for the broader connections. These findings resonate with the existing integrative reminiscence practice [3] and directly motivated the staged design of MemoryWeave: revisit first, then relate.

In this study, MemoryWeave addresses several key design considerations. First, rather than freely generating reflective content, the pipeline extracts structured context from each memory before generating any prompt, keeping all outputs anchored in the user's own material. Second, rather than imposing a detected resonance, users choose which cross-memory connection to explore, preserving their sense of control over the dialogue. Finally, all prompts and dialogue turns are explicitly non-directive, avoiding diagnostic claims, advice-giving, and unsupported interpretations.

In future work, we plan to evaluate MemoryWeave with a within-subjects study comparing two conditions. In Condition A (baseline), participants completed six AI-guided single-memory micro-reflections using the savoring prompts from Stage 1, with no cross-memory activity following. In Condition B (full pipeline), participants completed the same six micro-reflections using a different memory subset, followed by the full Stage 2 cross-memory dialogue. This comparison directly characterizes what the staged pipeline affords beyond the simpler single-memory baseline. Prior to both sessions, participants uploaded their personal photo-text memories to their archive. Data collection included micro-reflection written responses, cross-memory dialogue transcripts, post-session questionnaires measuring perceived reflection depth, personal insight, and reflective-emotional engagement, and a semi-structured end-of-study interview in which participants compared both conditions from direct experience. The central research question is: does the

staged MemoryWeave pipeline, which combines single-memory reminiscence reflection with AI-facilitated cross-memory meaning-making, support deeper reflection and greater personal insight than single-memory reflection alone?

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